南方科技大学
SUSTech

光学材料的纳米力学行为与加工缺陷衍化研究

理为源 工为干 需为导

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南方科技大学

SZKL·SIMI

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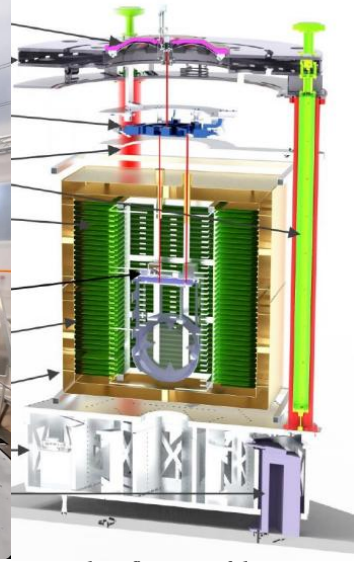
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一、研究背景及意义

深空探测——X射线空间望远镜（单晶硅）

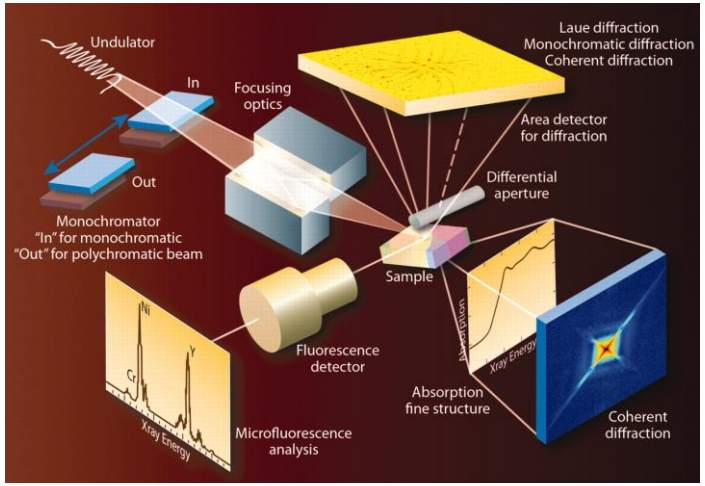


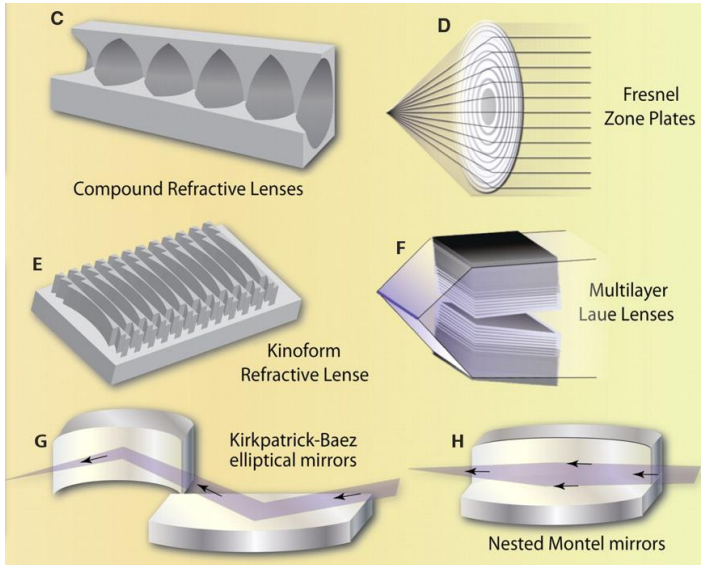




Castelvecchi, D. (2019). Space telescope to chart first map of the Universe in high-energy X-rays. Nature, 570(7760), 149–150.
Amorosi, A. (2024). E-Test Seismic Isolation. ETpathfinder Workshop January 2024, Maastricht University.

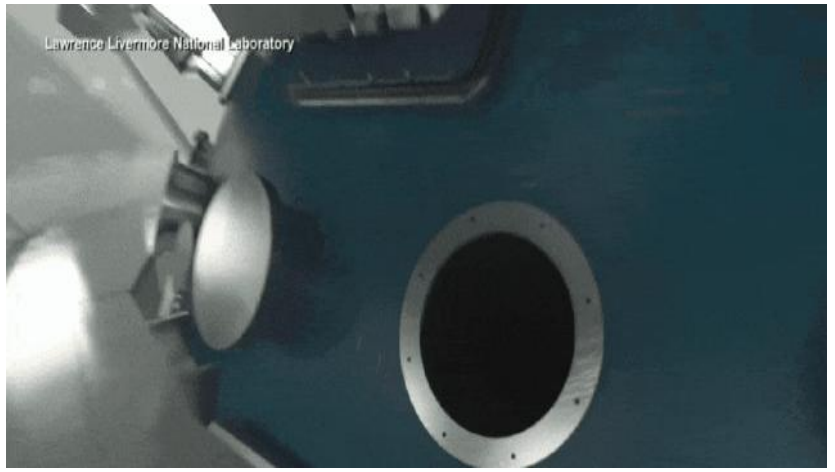
同步辐射——X射线反射镜（单晶硅）

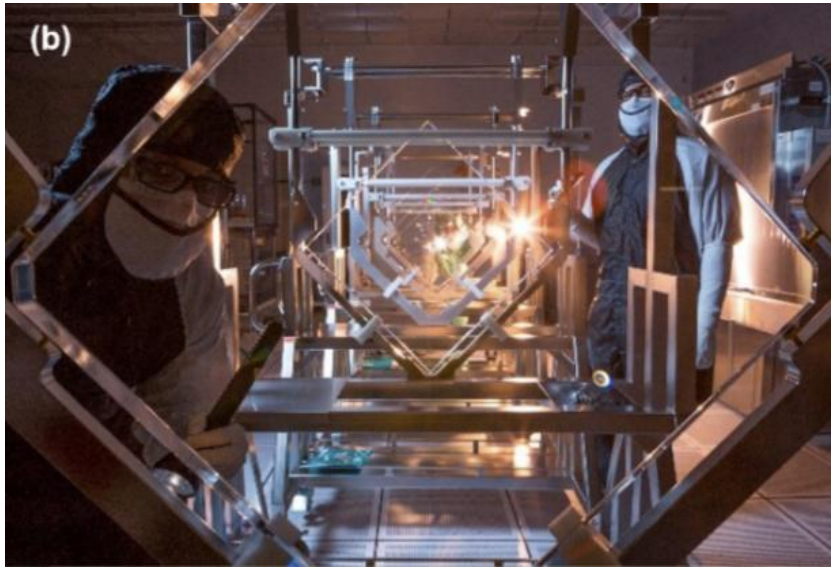




Ice, G. E., Budai, J. D., & Pang, J. W. L. (2011). The Race to X-ray Microbeam and Nanobeam Science. Science, 334(6060), 1234–1239.

惯性约束核聚变点火装置——高性能透镜（融石英）





Moore, L. A., & Smith, C. M. (2022). Fused silica as an optical material [Invited]. Optical Materials Express, 12(8), 3043.

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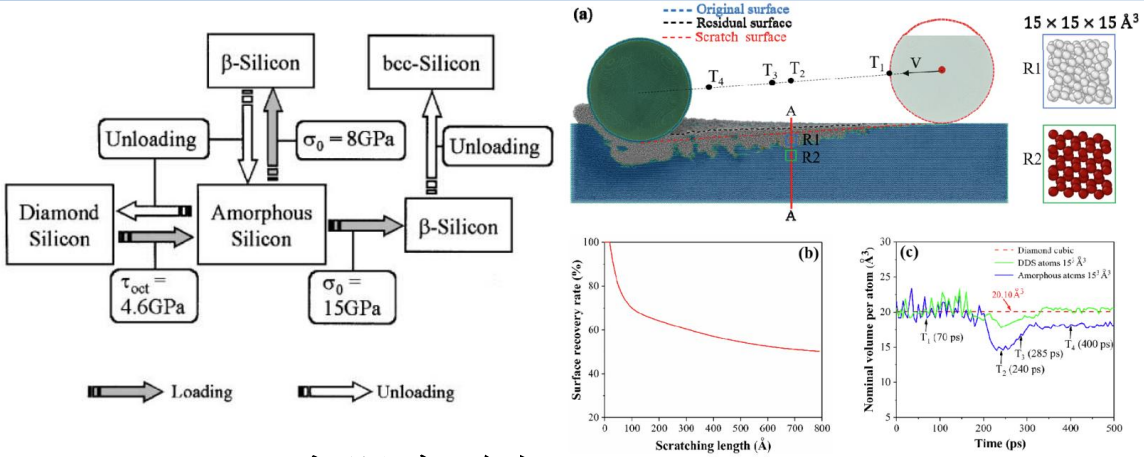
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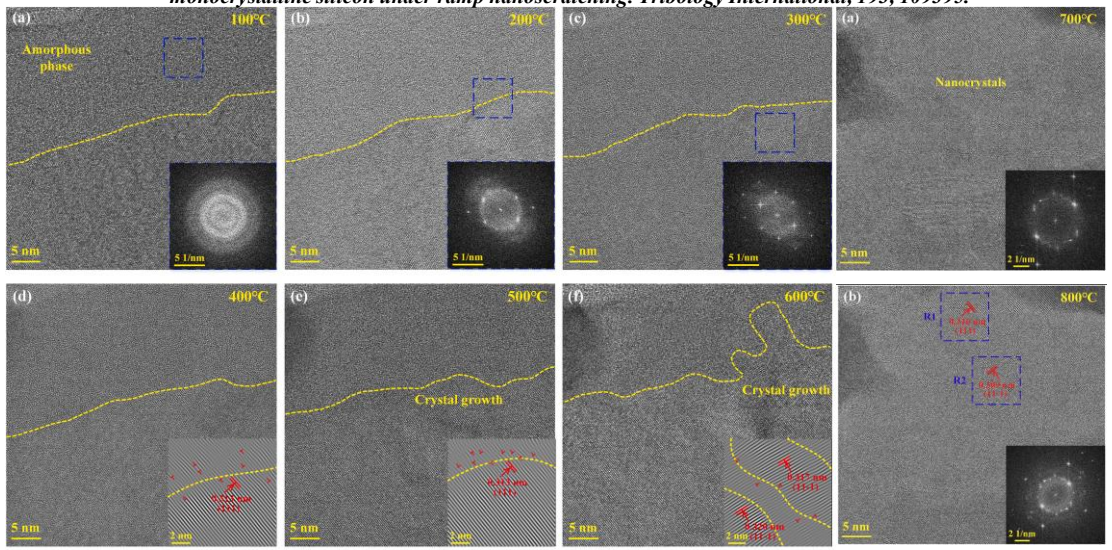
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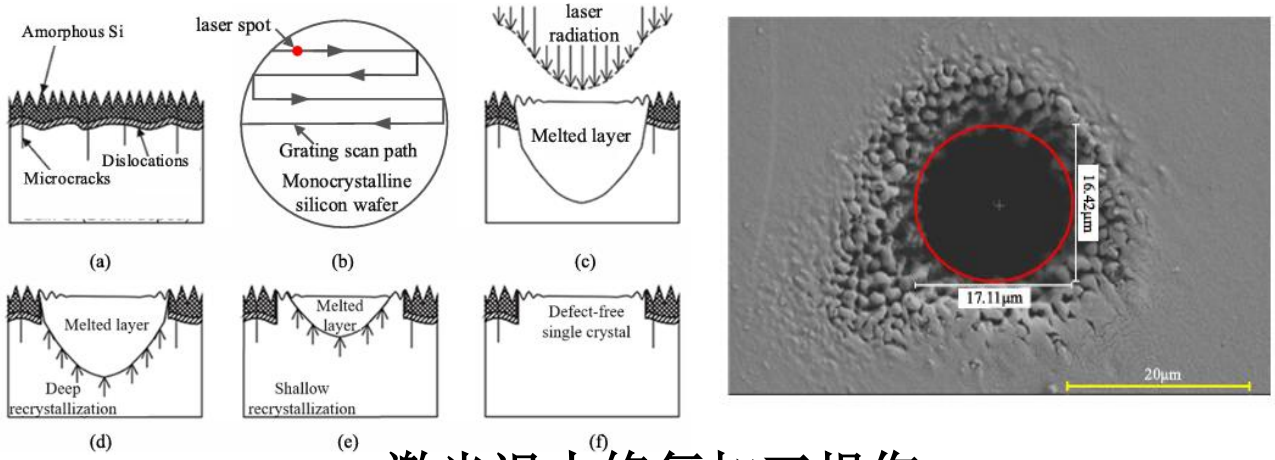
室温实验与分子动力学分析

Zhang, L., & Zarudi, I. (2001). Towards a deeper understanding of plastic deformation in mono-crystalline silicon. International Journal of Mechanical Sciences, 43(9), 1985～1996.
Li, Z., Li, Y., & Zhang, L. (2024). On the deformation mechanism and dislocations evolution in monocrystalline silicon under ramp nanoscratching. Tribology International, 193, 109395.



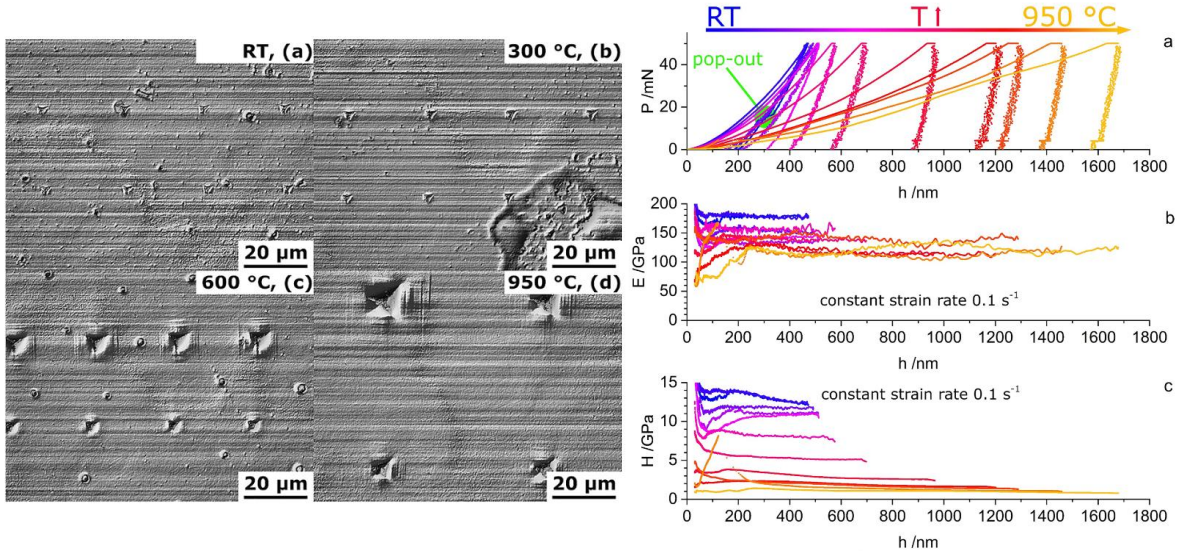
高温原位透射电镜实验

Li, Z., & Zhang, L. (2024). In-situ TEM investigation of dislocation healing and recrystallization in nanoscratched silicon at elevated temperatures up to 800 ° C. Journal of Materials Research and Technology



激光退火修复加工损伤

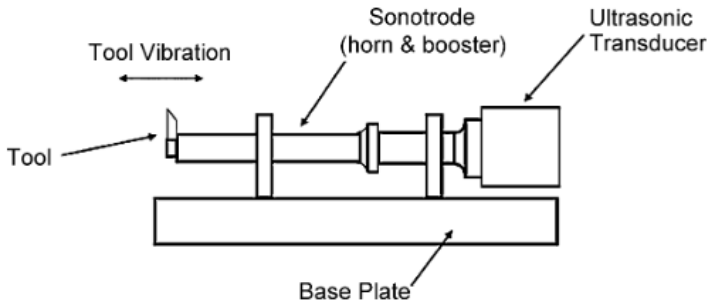
Sun, G., Deng, C., Hao, W., Zhao, J., Fan, M., Ding, J., & Ji, X. (2025). Study on mechanism and characteristics of laser repair of surface damage after ultra-precision grinding of monocrystalline silicon. Optics & Laser Technology, 186, 112660.

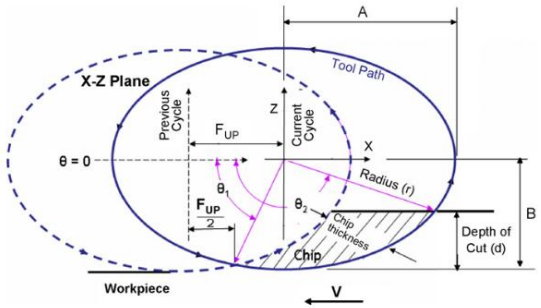


高温压痕实验

Schaffar, G. J. K., Tscharnuter, D., & Maier–Kiener, V. (2023). Exploring the high-temperature deformation behavior of monocrystalline silicon – An advanced nanoindentation study. Materials & Design

二、研究现状





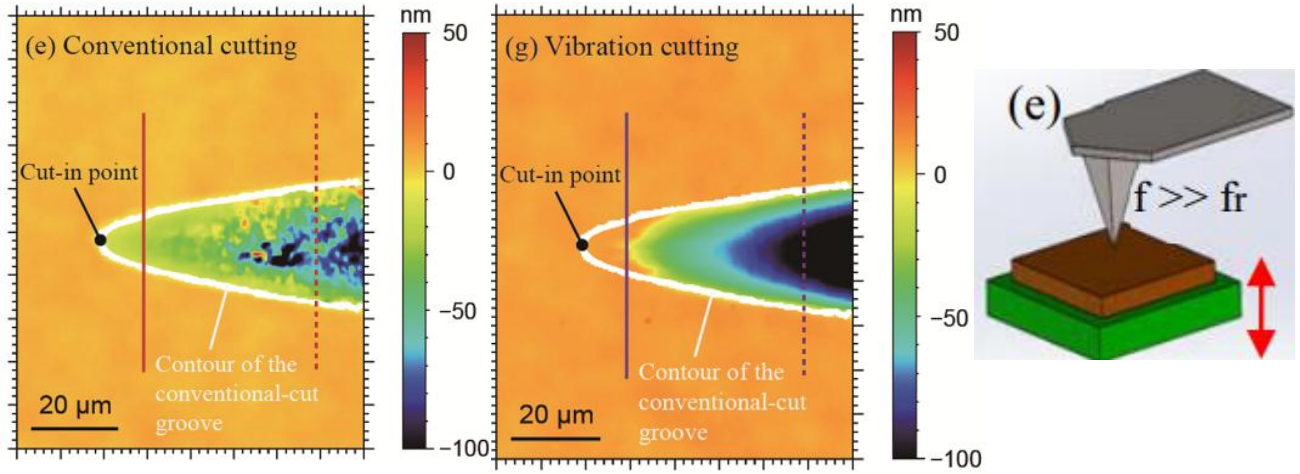
Material		Process	Tool nose radius (mm)	Depth of cut (μm)	Surface roughness (μm)	
Work	Tool				VAM	Conventional
SS	Carbide	1D	0.20	50	4.3 PV	20.0 PV
Inconel	Carbide	1D	0.20	50	4.5 PV	60.0 PV
Al-SiC[b]	SCD	1D	0.80	1	0.578 PV	0.859 PV
Fused silica	SCD	1D	0.80	2	0.100 Ra	0.260 Ra
CFRP[c]	SCD	1D	0.50	200	2.80 Ra	4.00 Ra
LY12 (aluminum)	PCD	1D	0.50	150	0.63 Ra	0.94 Ra
304 SS	PCD	1D	0.20	20	0.152 Ra	0.239 Ra
PMMA (CR-39)	SCD	1D	NR	2	0.1 Ra	0.2 Ra
Steel	SCD	2D	1.00	3.5	0.04 PV	0.52 PV
Brass	SCD	2D	NR	1	0.04 PV	1.00 PV
Inconel	Carbide	1D	NR	800	0.284 Ra	0.510 Ra

一维、二维振动辅助加工刀具轨迹示意图

Yang, Z., Zhu, L., Zhang, G., Ni, C., & Lin, B. (2020). Review of ultrasonic vibration-assisted machining in advanced materials. International Journal of Machine Tools and Manufacture, 156, 103594.

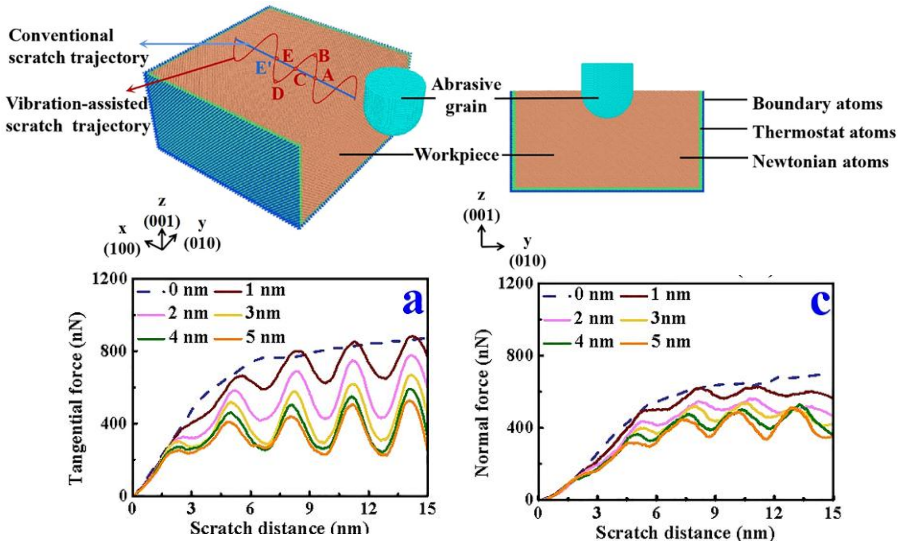
文献中报道振动辅助加工效果对比

Brehl, D. E., & Dow, T. A. (2008). Review of vibration-assisted machining. Precision Engineering, 32(3), 153～172.



垂向振动辅助加工单晶硅实验

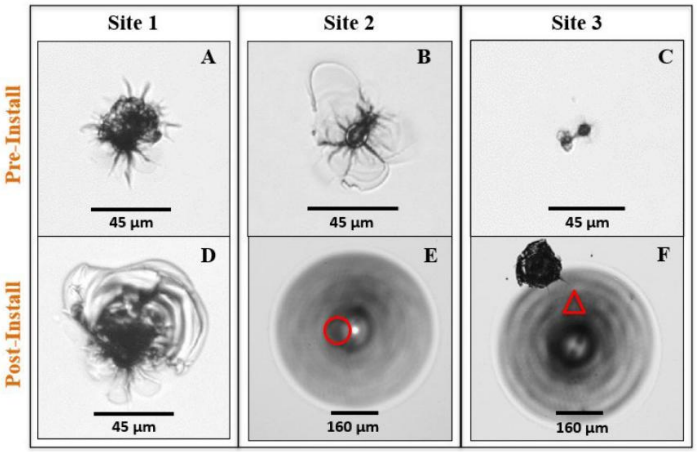
Huang, W., & Yan, J. (2023). Towards understanding the mechanism of vibration-assisted cutting of monocrystalline silicon by cyclic nanoindentation. Journal of Materials Processing Technology, 311, 117797.
Deng, J., Dong, J., & Cohen, P. (2016). High Rate 3D Nanofabrication by AFM-based Ultrasonic Vibration Assisted Nanomachining. Procedia Manufacturing, 5, 1283～1294.



分子动力学模拟探究振动辅助加工单晶硅

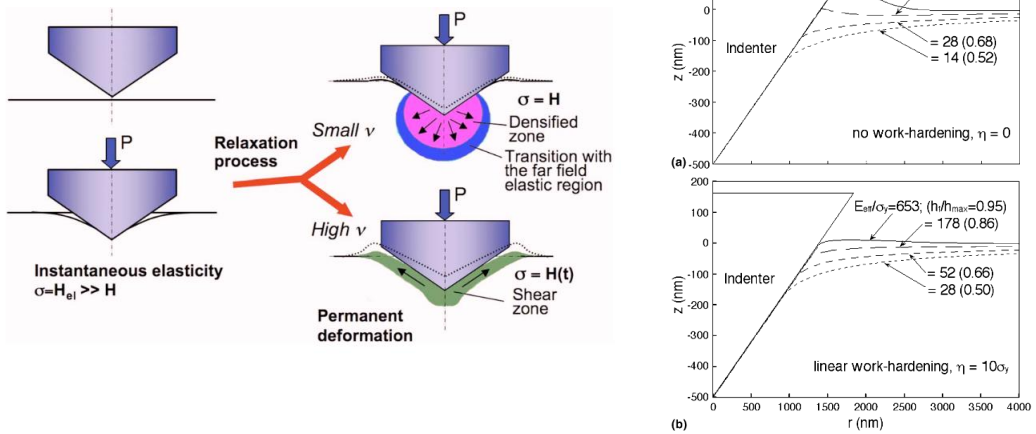
Chen, Y., Hu, Z., Jin, J., Li, L., Yu, Y., Peng, Q., & Xu, X. (2021). Molecular dynamics simulations of scratching characteristics in vibration-assisted nano-scratch of single-crystal silicon. Applied Surface Science, 551, 149451.

二、研究现状



缺陷导致辐射损伤

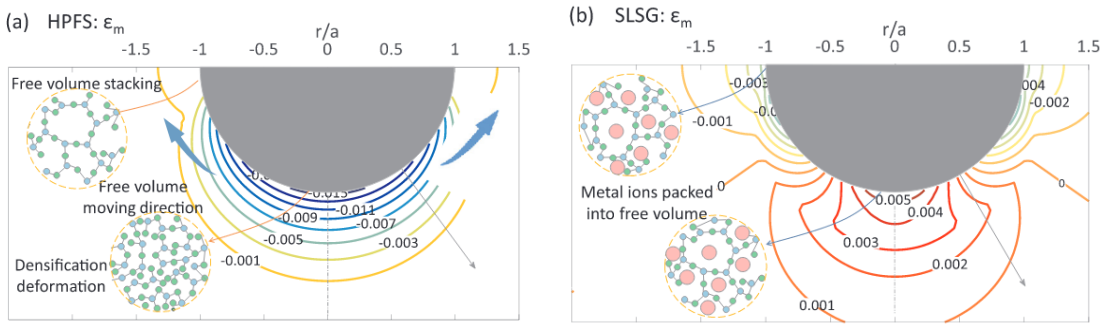
Palma, K., Raman, R. N., Miller, C. F., Trummer, S., Cross, D. A., & Carr, C. W. (2023). Repair strategies for fused silica wedged focus lens with high density filamentary damage. Laser-Induced Damage in Optical Materials 2023 (p. 26). SPIE.



玻璃材料粘性流动与致密化理论模型

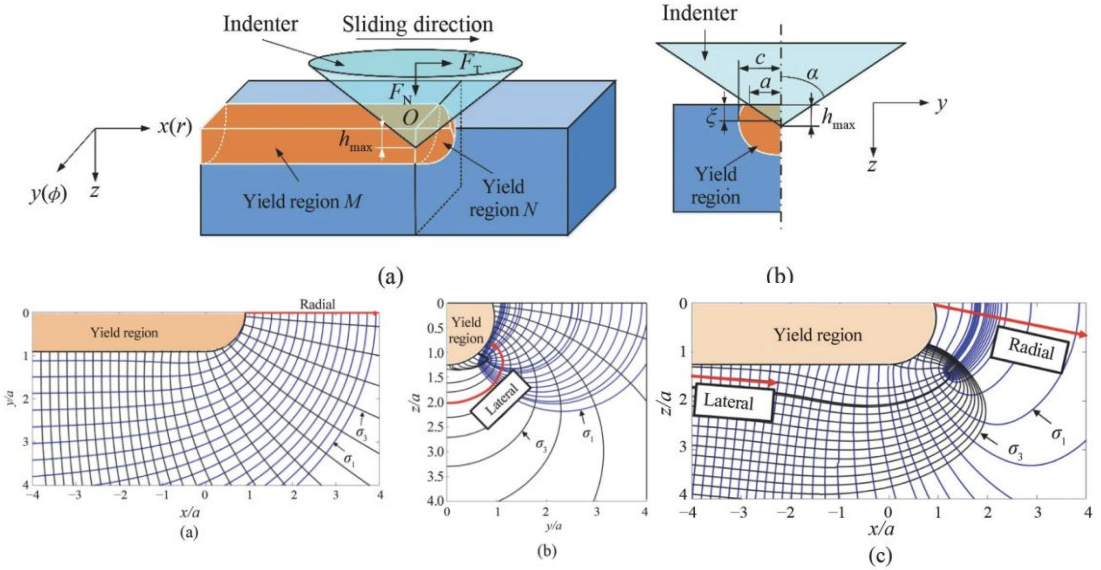
Oliver, W. C., & Pharr, G. M. (2004). Measurement of hardness and elastic modulus by instrumented indentation: Advances in understanding and refinements to methodology. Journal of Materials Research, 19(1), 3–20.

Rouxel, T., Ji, H., Guin, J. P., Augereau, F., & Rufflé, B. (2010). Indentation deformation mechanism in glass: Densification versus shear flow. Journal of Applied Physics, 107(9), 094903.



微观结构致密化分析

Wang, W., Li, Z., Yao, P., Li, J., Chen, F., & Liu, Y. (2020). Sink-in/pile-up formation and crack nucleation mechanisms of high purity fused silica and soda-lime silica glass during nanoindentation experiments. Ceramics International, 46(15), 24698–24709.



应力场模型分析

Li, C.-S., Zhao, N., Zhang, L.-C., Ding, J.-J., Sun, L., Duan, D.-Z., Kang, C.-W., & Jiang, Z.-D. (2023). An analytical method for assessing the initiation and interaction of cracks in fused silica subjected to contact sliding. Advances in Manufacturing.

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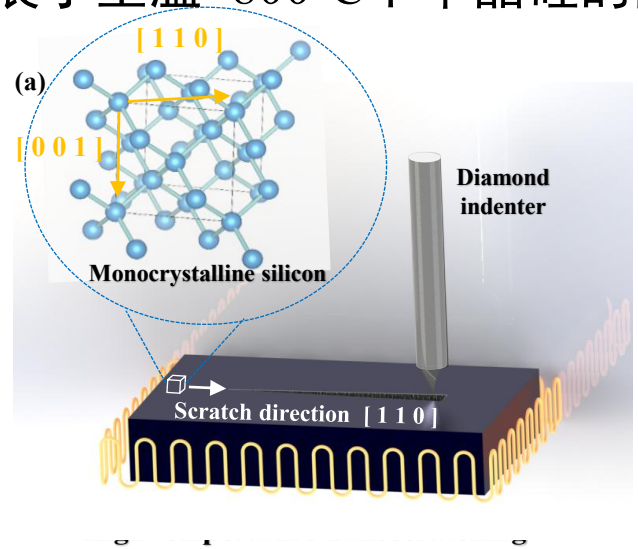
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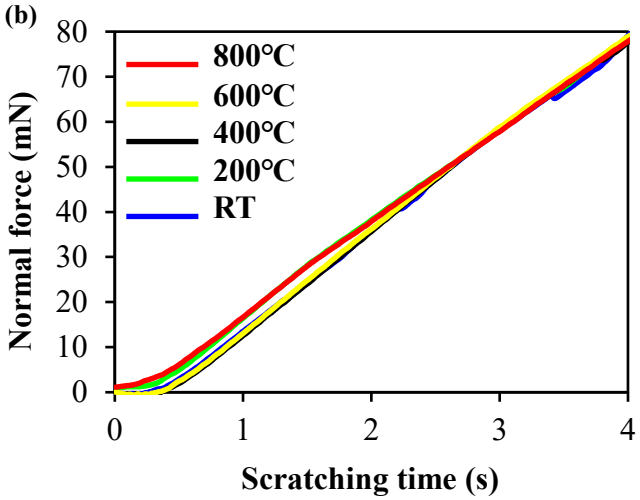
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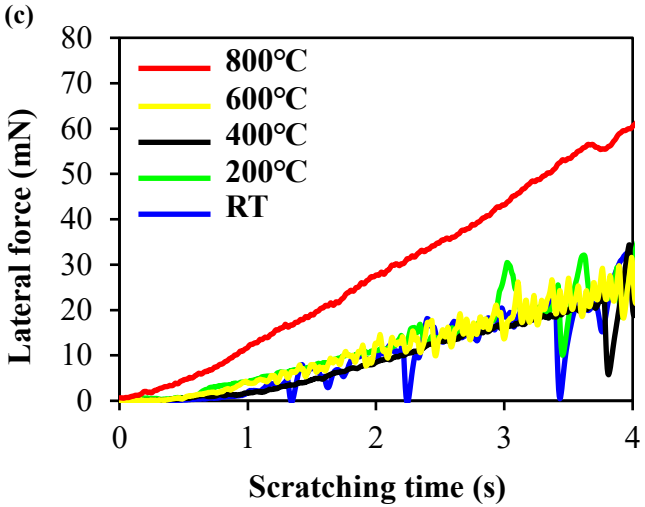
- 开展了室温~800℃下单晶硅的高温纳米划刻加工实验，研究应力场与温度场耦合下的材料去除机理；



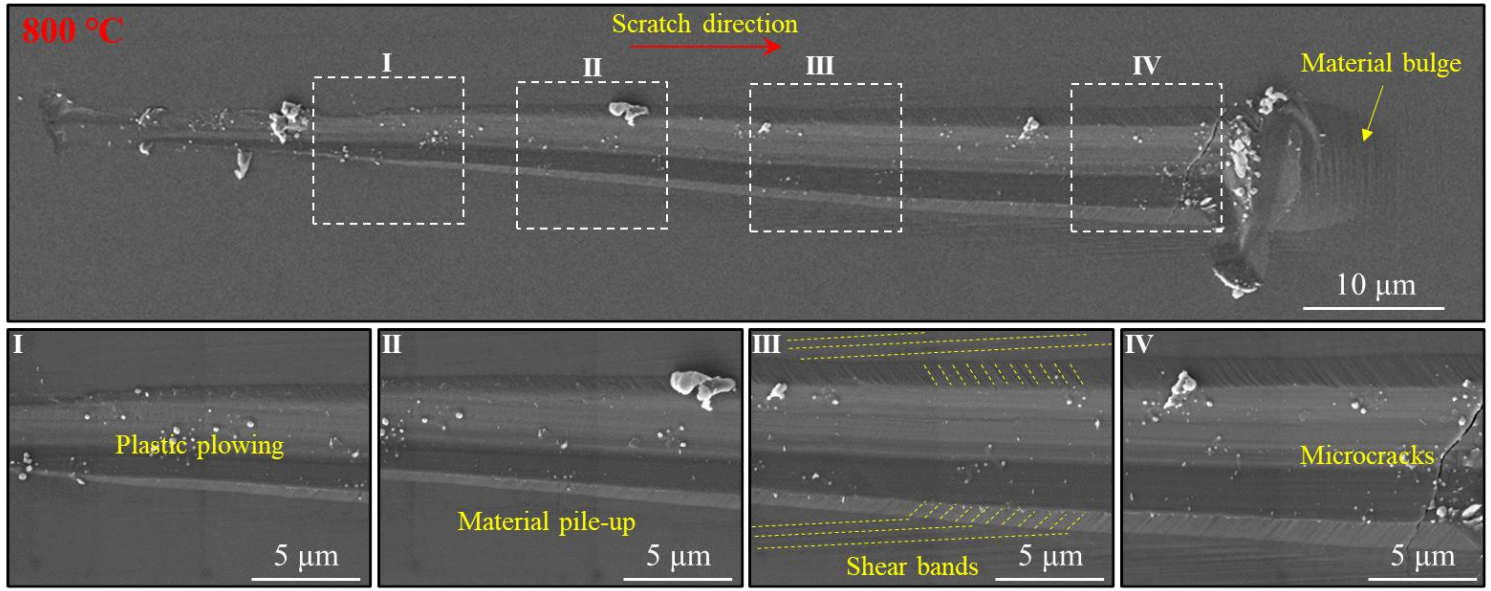
（RT~800℃）



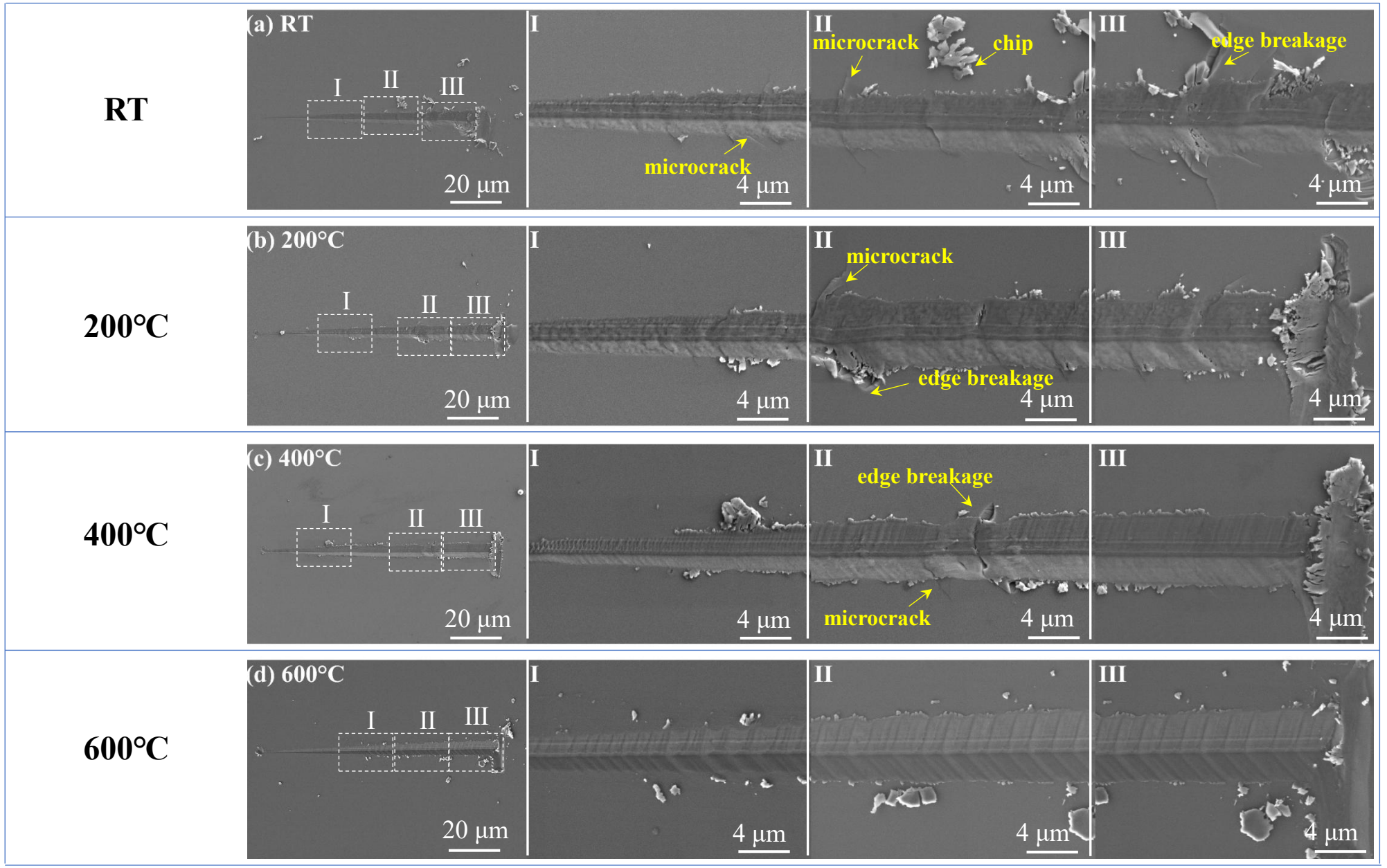
Scratching Normal Force



Scratching Lateral Force



三、热-力耦合场下单晶硅的纳米力学行为与缺陷衍化机制



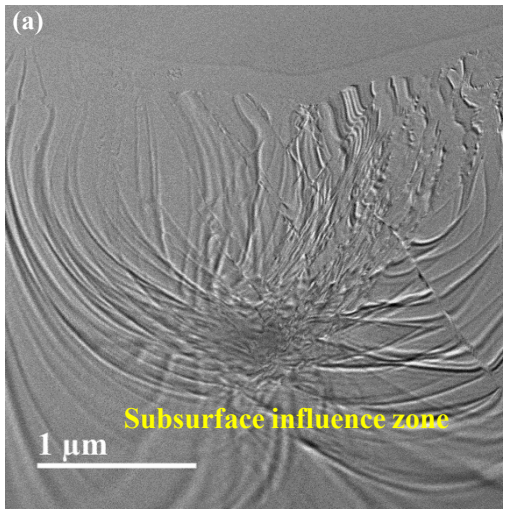
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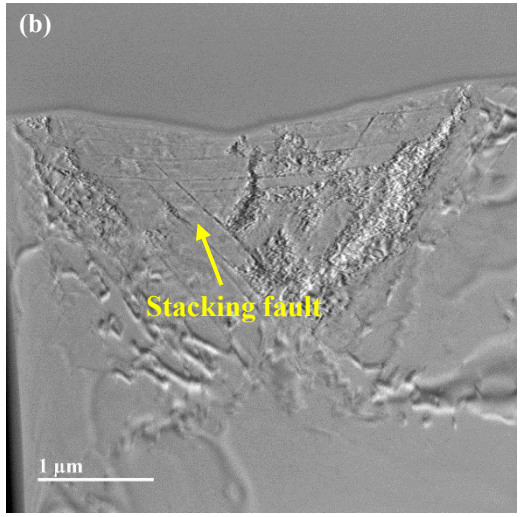
对室温、600℃、800℃划痕下方亚表面损伤层进行微观分析。综合发现600℃下单晶硅加工损伤较少。

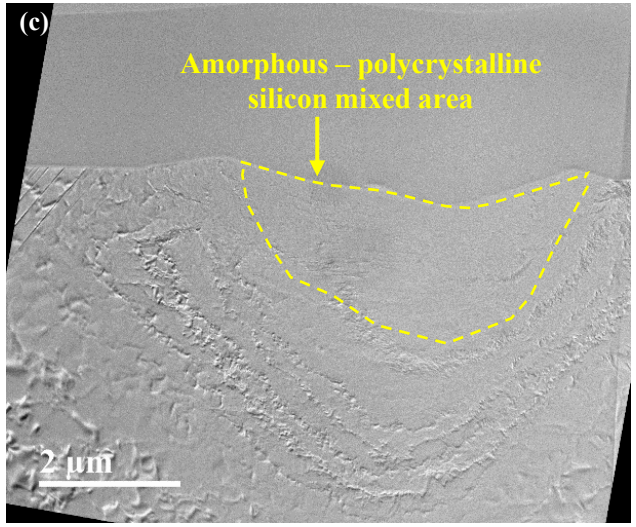
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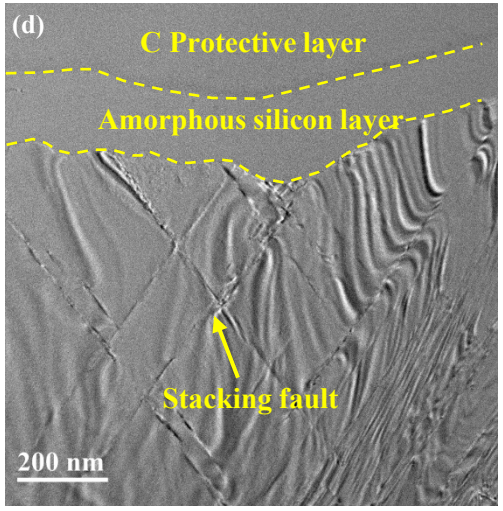
600℃

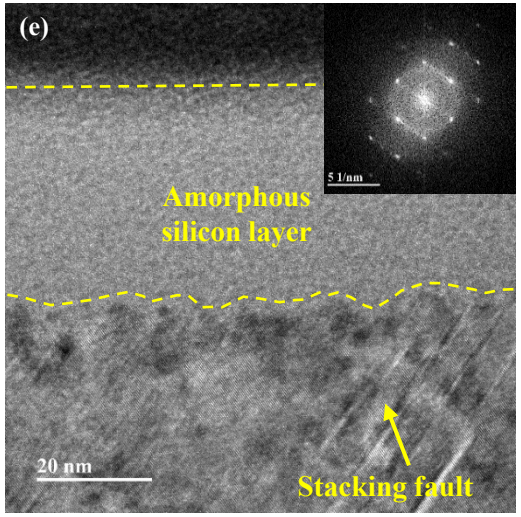
800℃

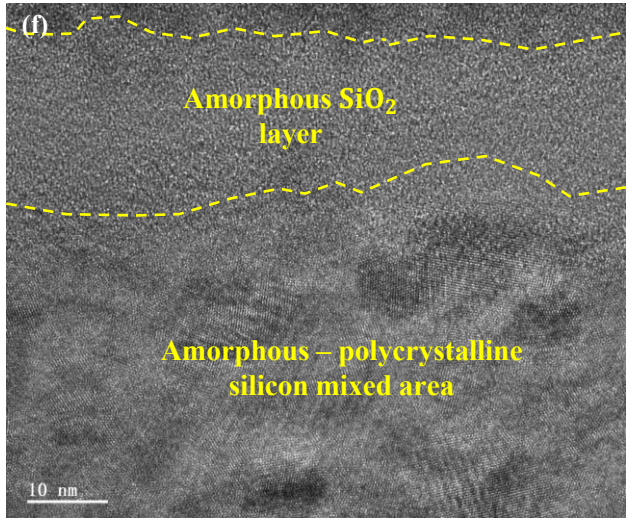












三、热-力耦合场下单晶硅的纳米力学行为与缺陷衍化机制

为探究不同温度下单晶硅加工缺陷衍化改变机理，开展分子动力学模拟研究。

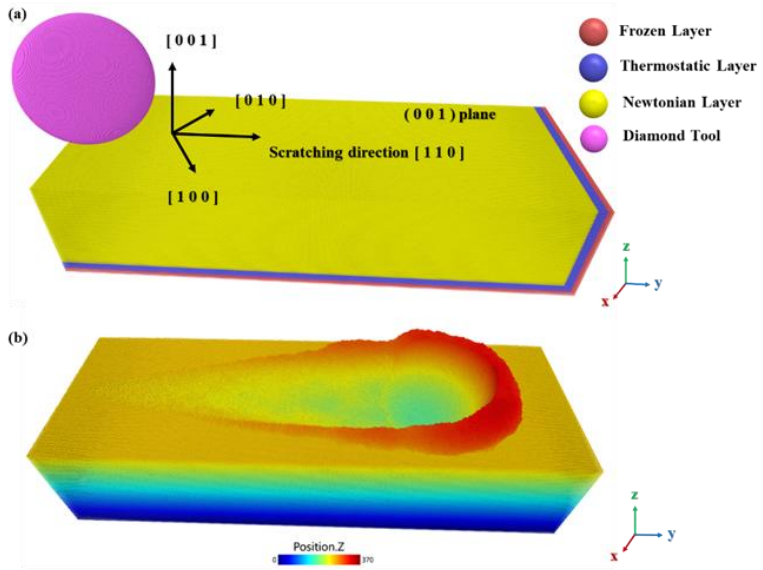
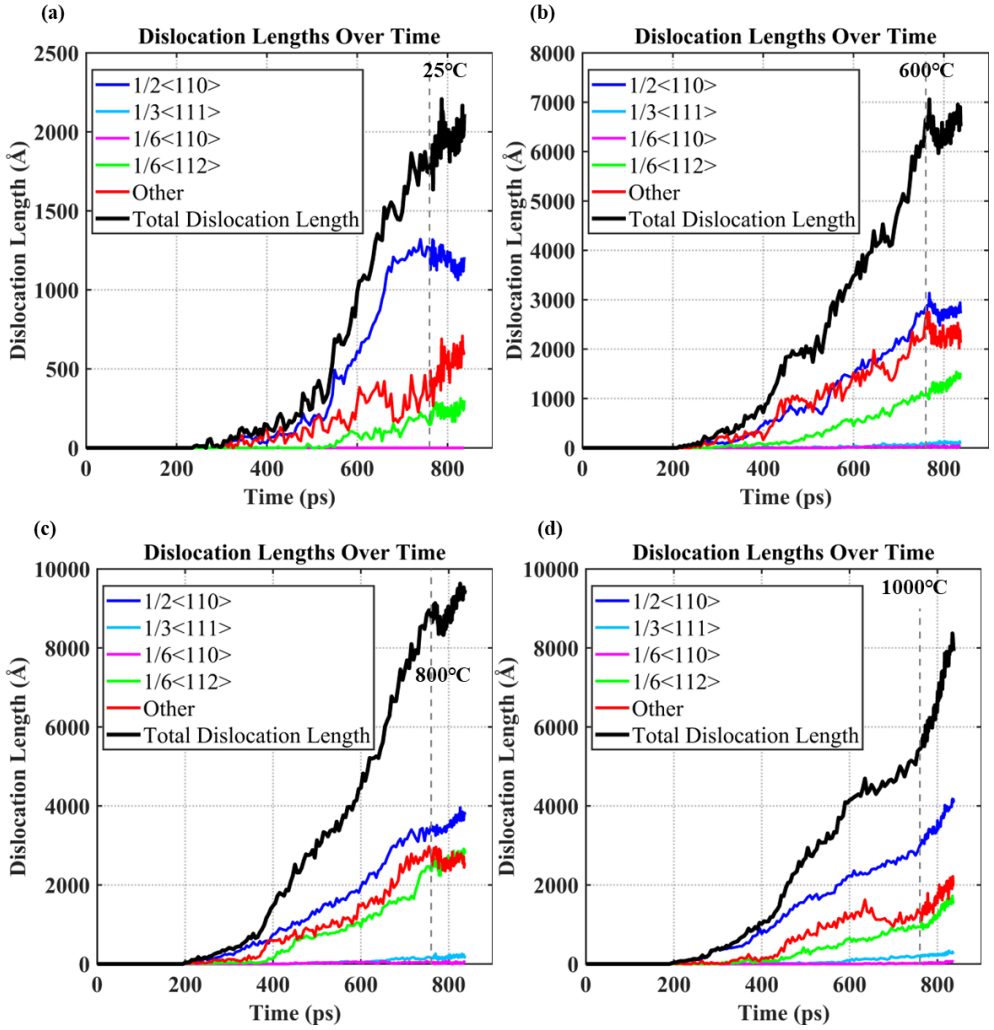


表 3-1 分子动力学模拟模型参数与势函数.

分子动力学仿真参数	
金刚石压头半径	14 nm
单晶硅块体尺寸	40.26 × 115.58 × 24.72 nm³
划痕方向速度	100 m/s
下压方向速度	10 m/s
Si-Si势函数	Erhart-Albe Tersoff
Si-C势函数	Morse



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振动辅助加工的实验设置与力响应结果。

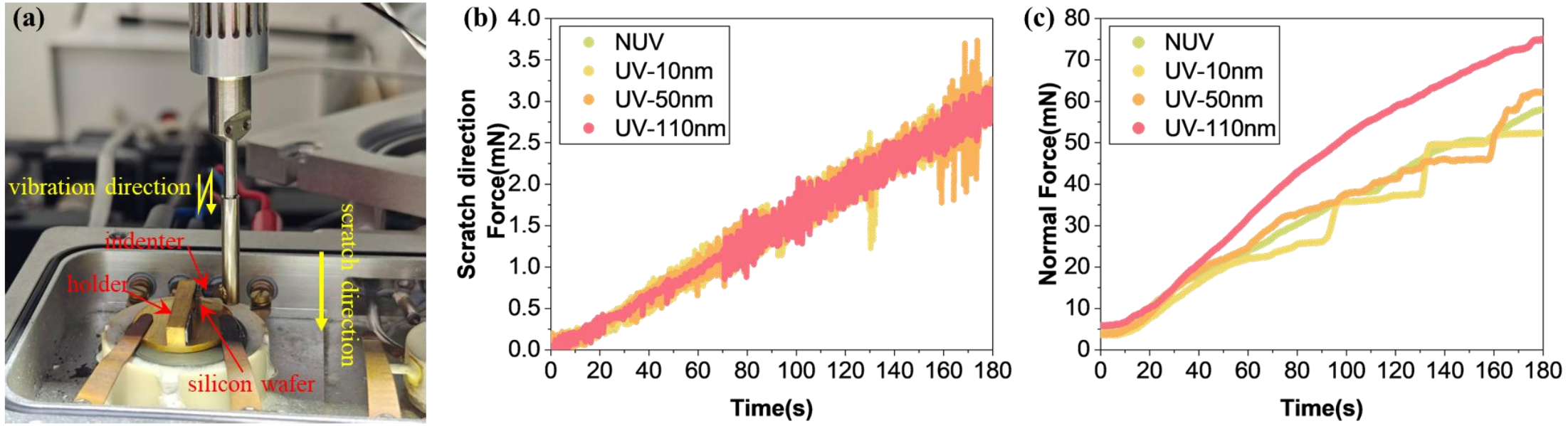
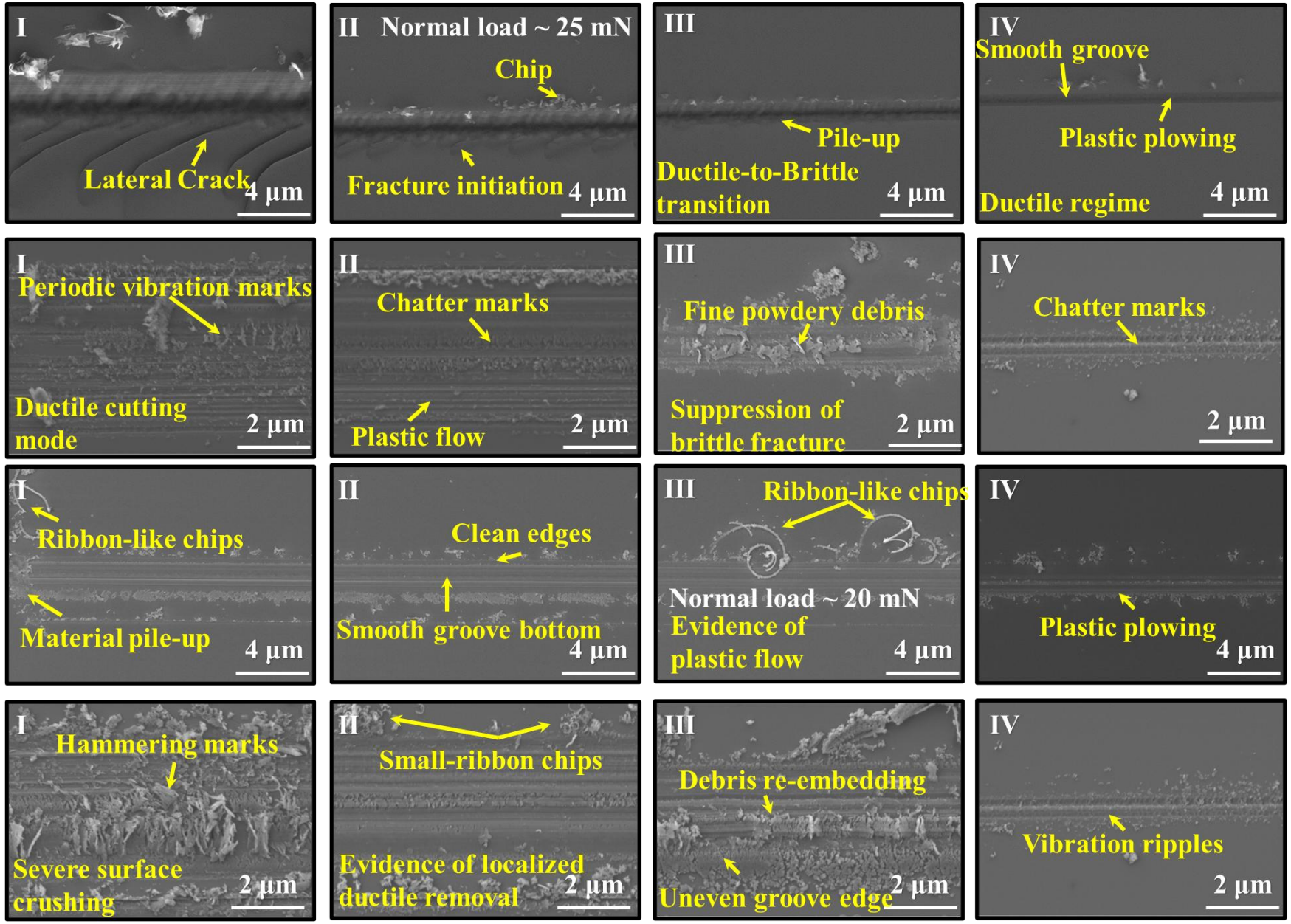


表 4-1 超声振动辅助划痕实验的实验参数.

实验组别	Scratch direction amplitude(nm)	Normal direction amplitude(nm)	振动频率
UV-10 nm	10	0	102.9kHz
UV-50 nm	50	10	102.9kHz
UV-110 nm	110	20	102.9kHz

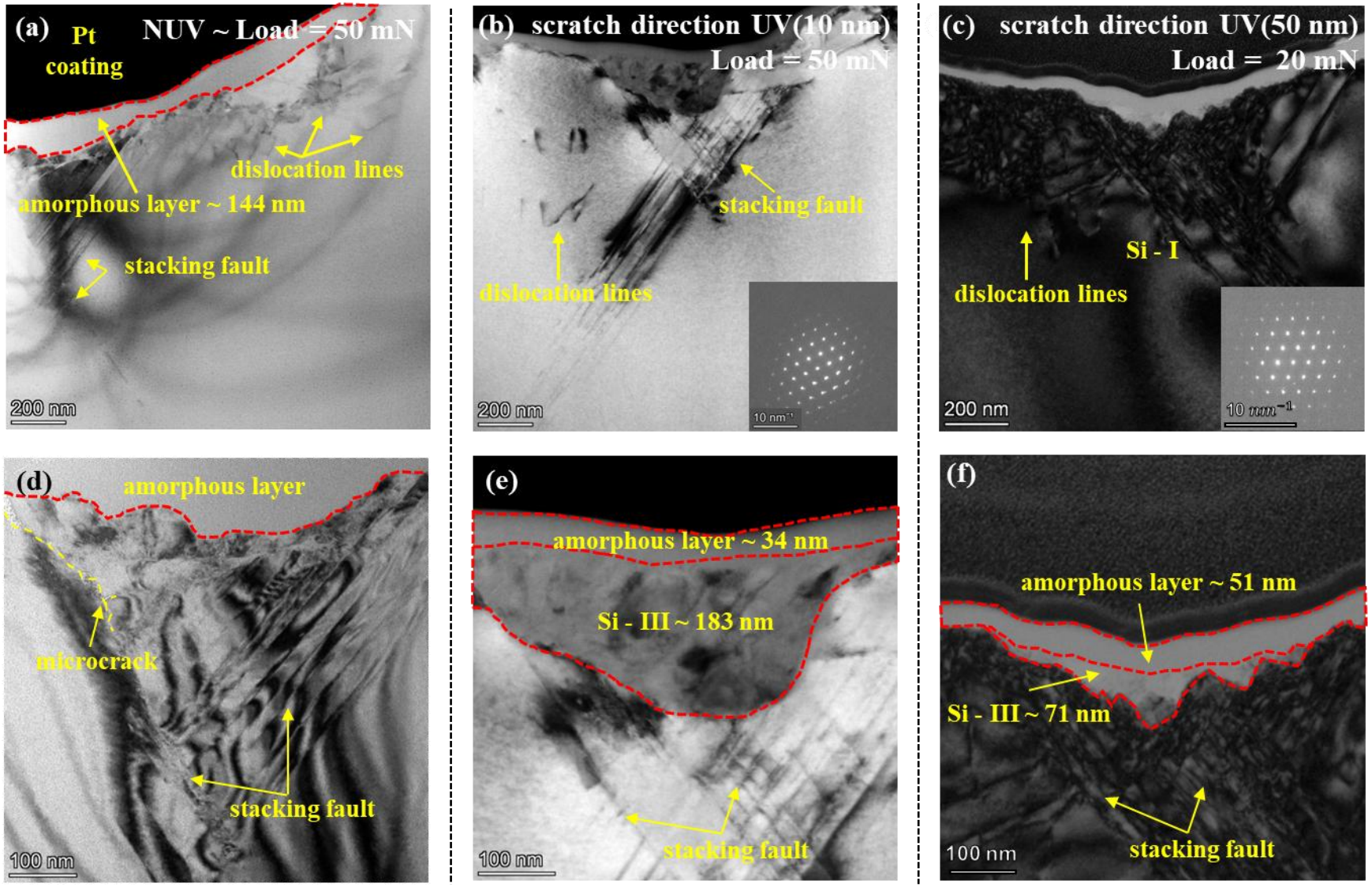
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四、超声振动场辅助下单晶硅的缺陷响应与亚稳态相变机理

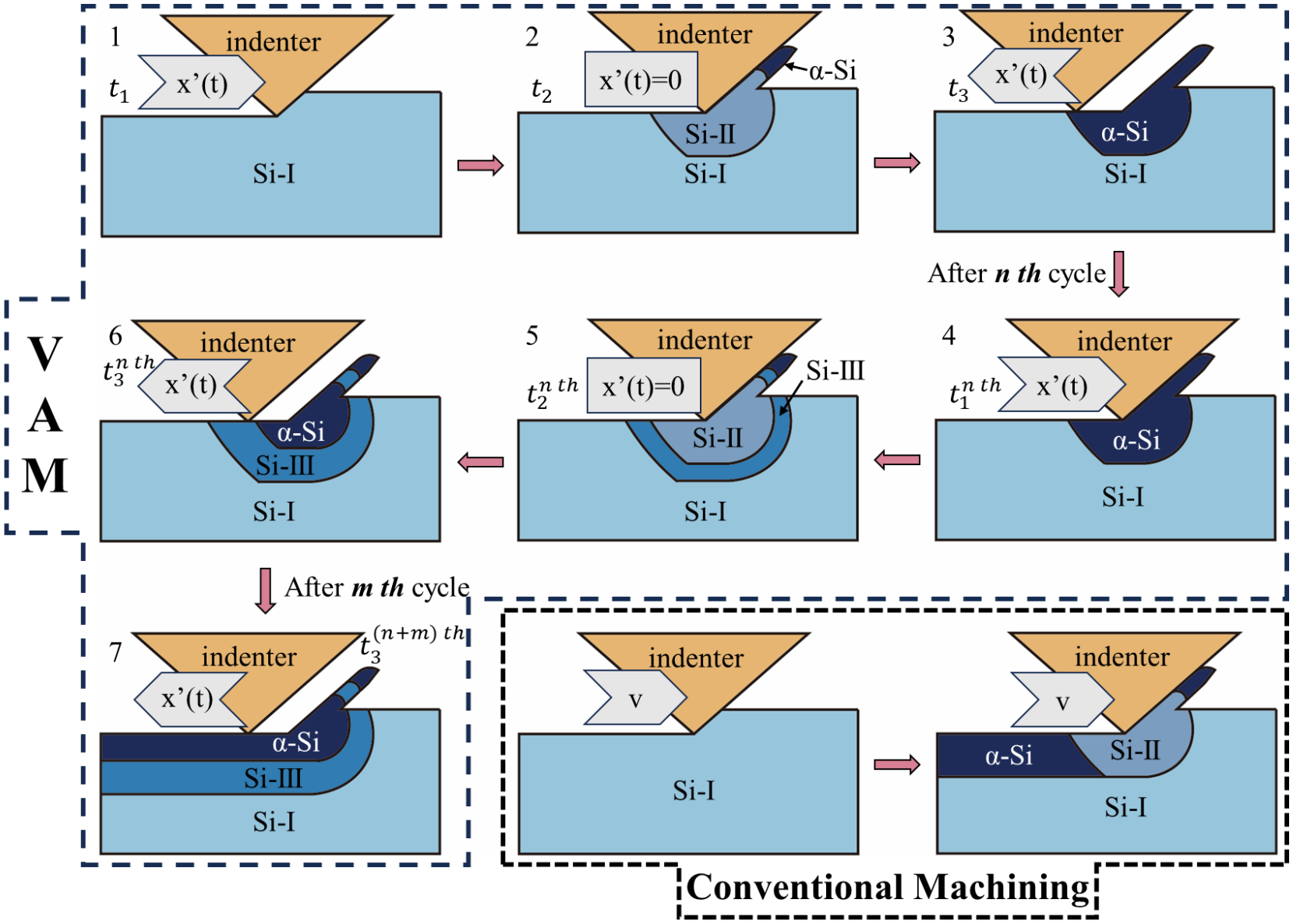
对不同工况下的亚表面损伤层进行分析，发现超声振动辅助加工可以显著减少加工导致的非晶层厚度。

NUV　　UV - 10 nm　　UV - 50 nm



四、超声振动场辅助下单晶硅的缺陷响应与亚稳态相变机理

与常规加工单晶硅相变路径相比，振动辅助加工单晶硅相变路径更为复杂，生成了大量亚稳相Si-III(BC8)相。



振动辅助加工与常规加工单晶硅相变机理示意图

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三、热-力耦合场下单晶硅的纳米力学行为与缺陷衍化机制

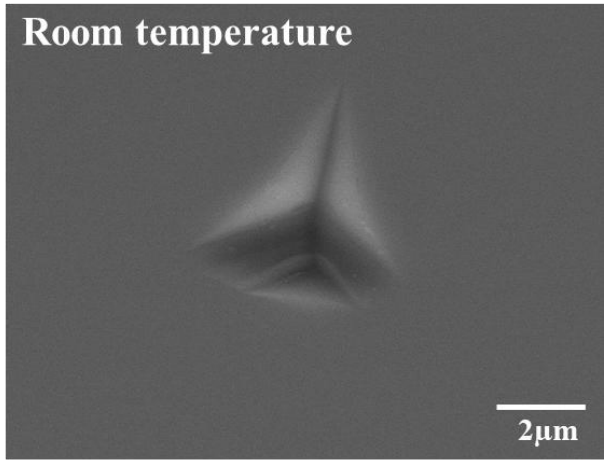
四、超声振动场辅助下单晶硅的缺陷响应与亚稳态相变机理

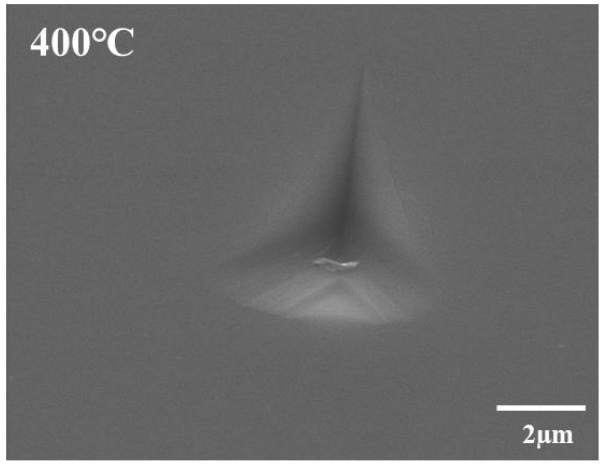
五、力-热场下融石英纳米尺度致密化表征与本构关系建立

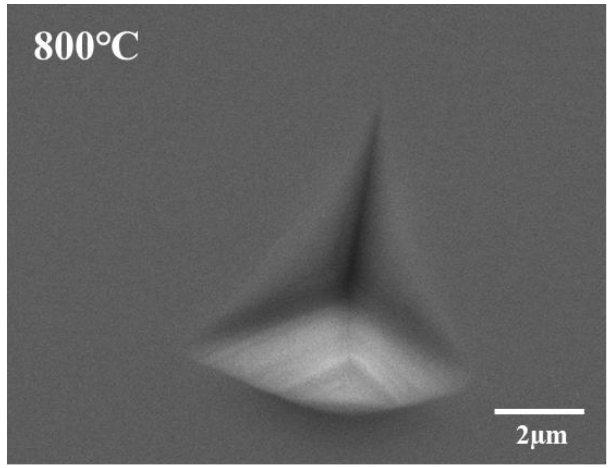
六、结论与展望

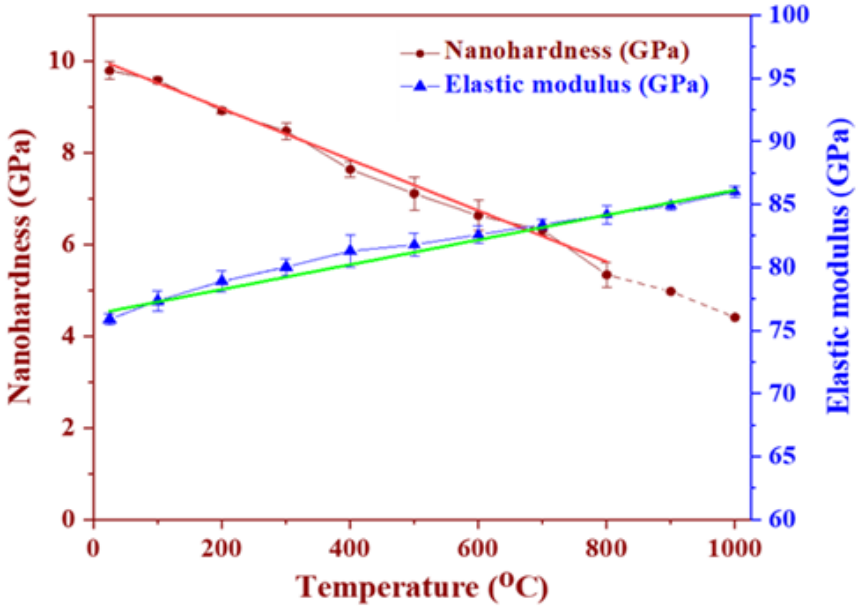
五、力-热场下融石英纳米尺度致密化表征与本构关系建立

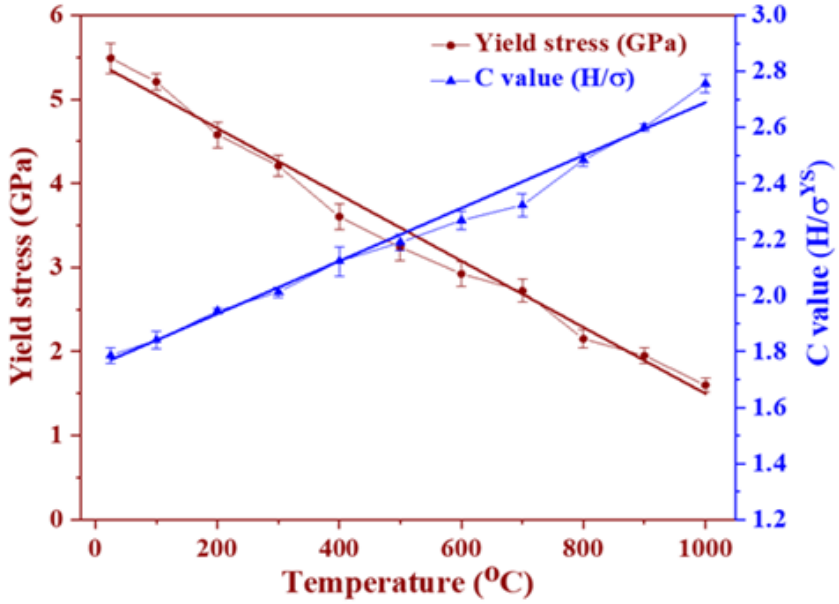
对不同温度下融石英进行纳米力学测试，测得材料常数。

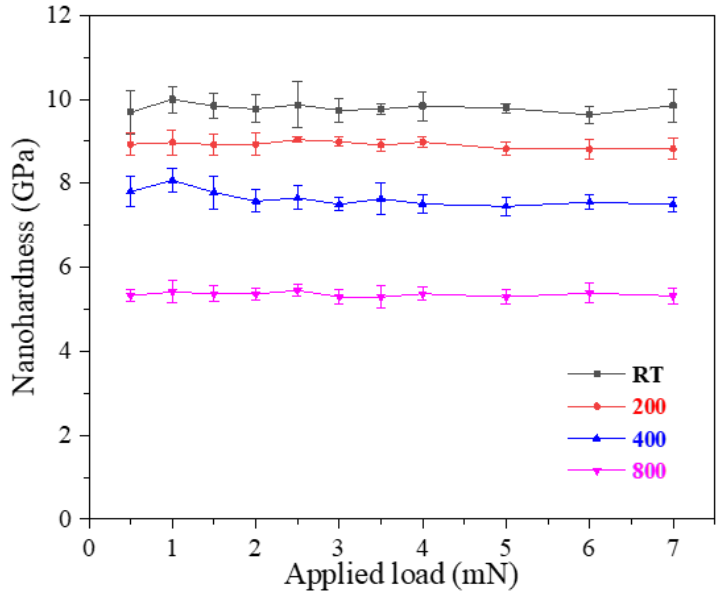




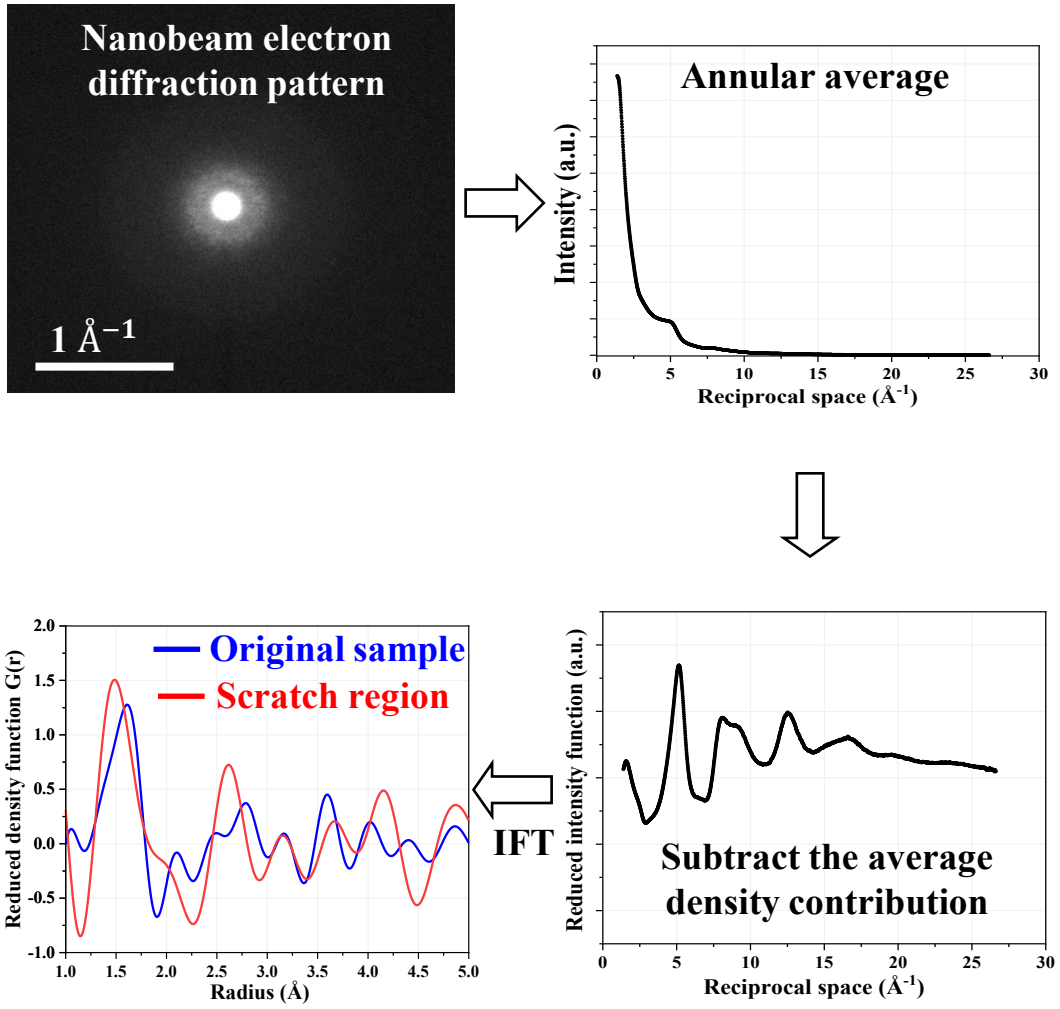




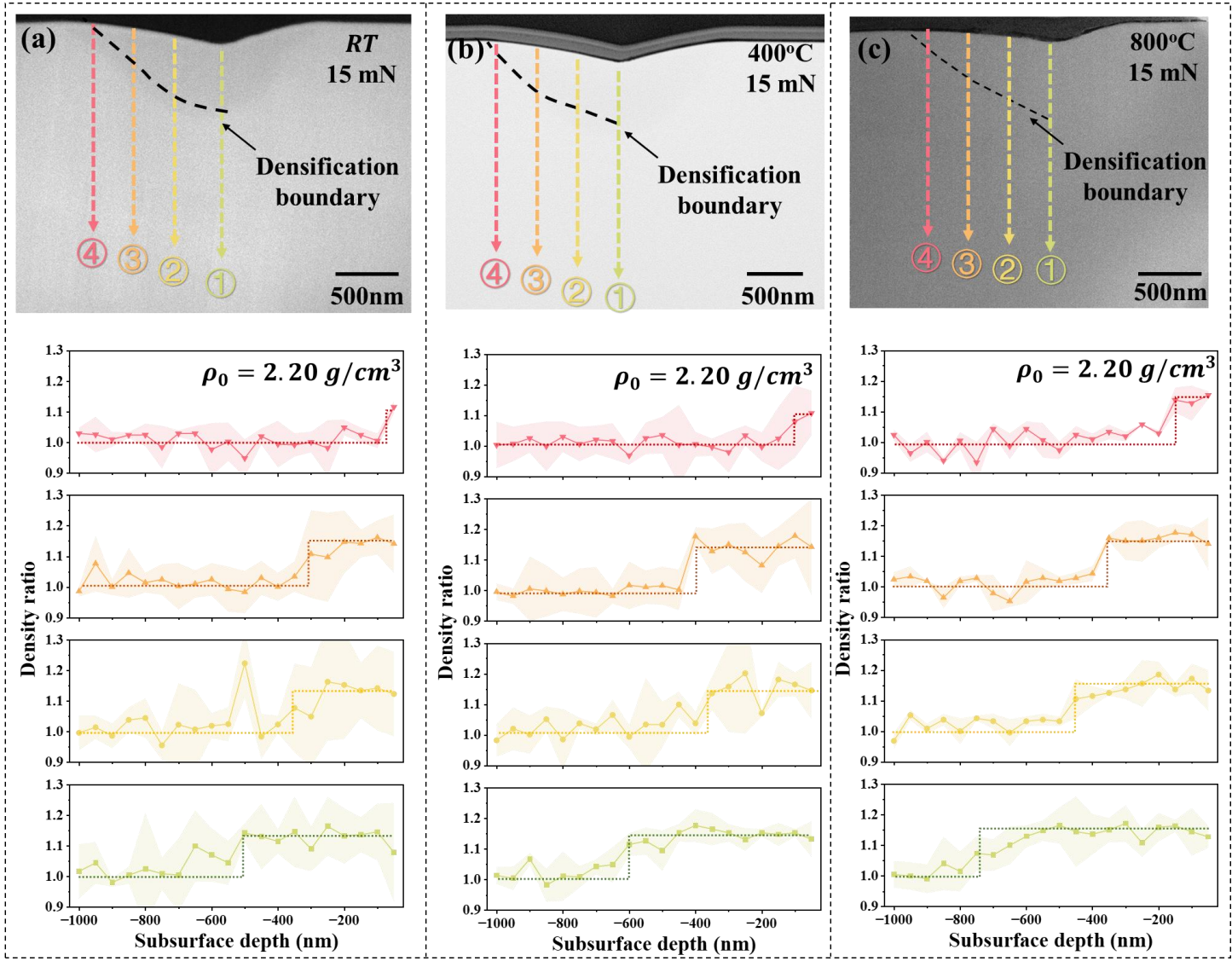




五、力-热场下融石英纳米尺度致密化表征与本构关系建立



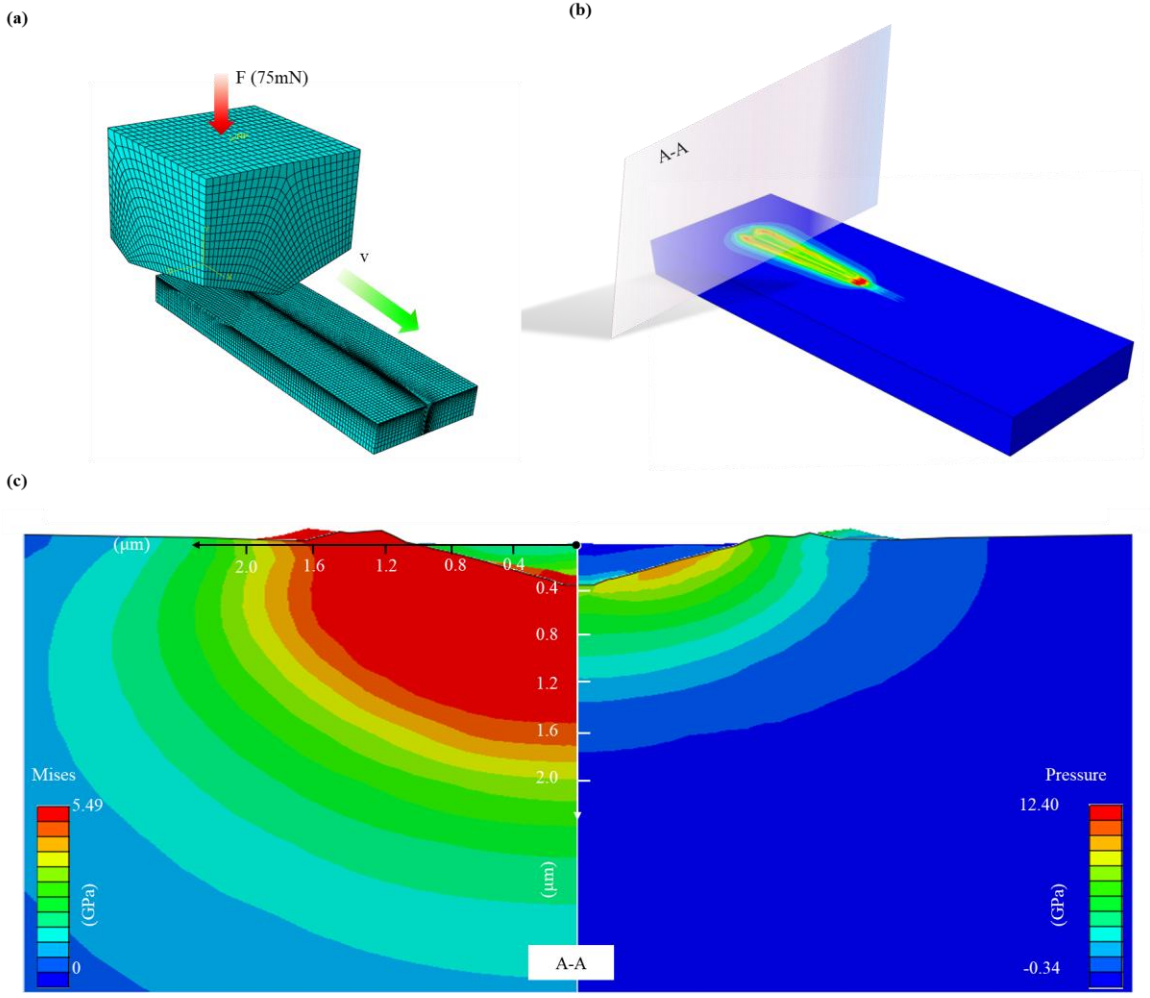
利用德拜方程从纳米束衍射斑获取约化密度函数的流程

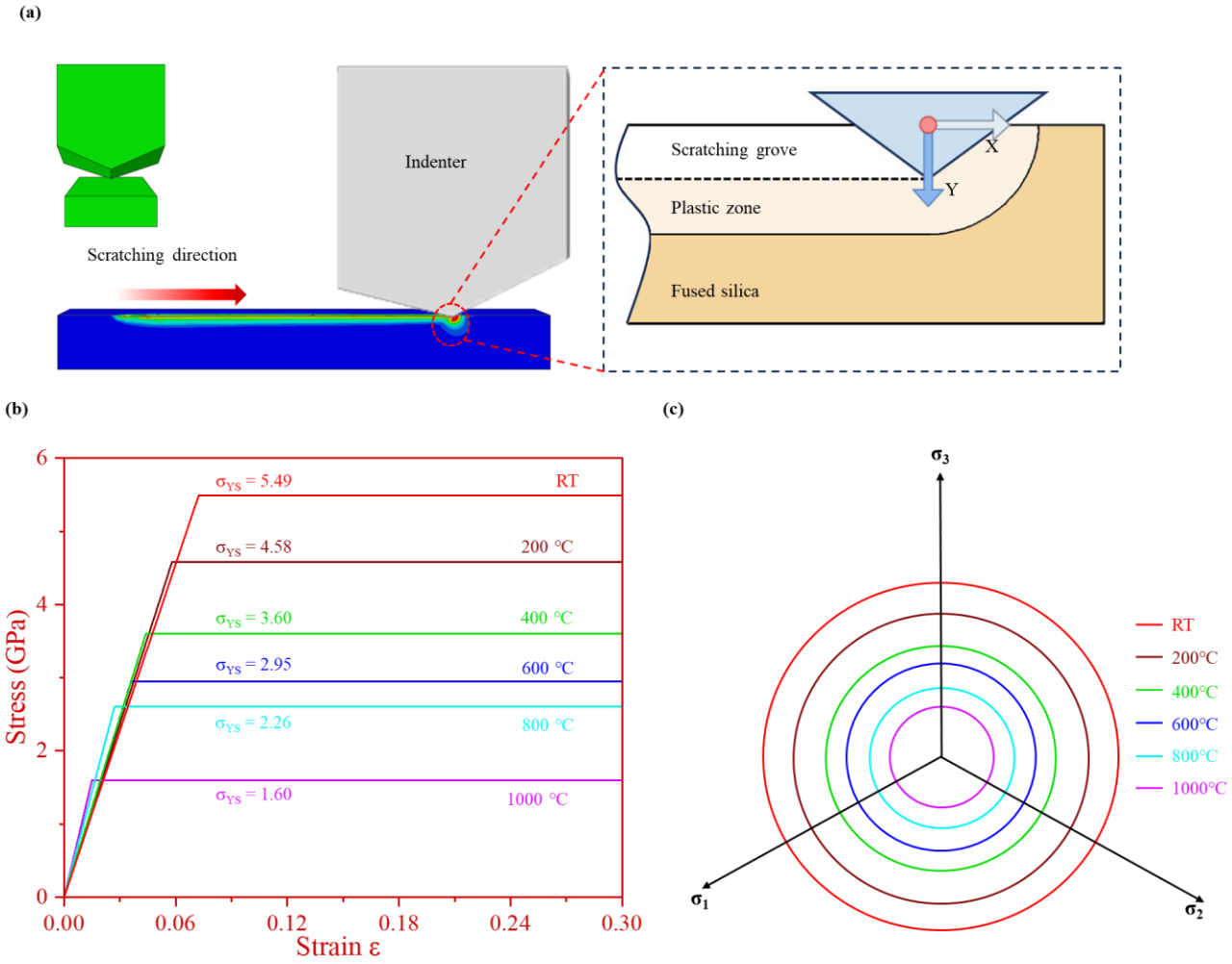


测得致密化结果的空间分布情况

五、力-热场下融石英纳米尺度致密化表征与本构关系建立

合作者有限元仿真结果以及融石英纳米尺度本构关系模型。





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六、结论与展望

六、结论与展望

主要研究结论

1. 明确 600℃ 为单晶硅热辅助加工的最佳温度窗口，该温度下亚表面非晶层厚度锐减至室温的 1/6。
2. 发现了超声振动辅助加工单晶硅能够抑制侧向裂纹提升塑性加工临界深度，并探究了亚表面相变机制。
3. 采用纳米束电子衍射技术在纳米尺度表征融石英亚表面致密化的分布，为证实融石英在复杂接触下的屈服行为遵循 von Mises准则，及其理想弹塑性本构框架提供了实验证明。

研究展望

1. 多能场复合加工技术的深度探索。
2. 从纳米划痕向宏观超精密加工的进阶。
3. 对融石英考虑实际加工时与水发生化学反应对其纳米力学响应的影响。

在校期间完成的相关学术成果

学术论文

[1]**Zhang M,** Li Z, Zhang L. Thermally Assisted Nanoscratching of Monocrystalline Silicon up to 800℃: Tribology and Subsurface Transformations. Wear. DOI: 10.1016/j.wear.2026.206707(已发表，IF=6.1，对应学术论文第3章)

[2]Zhang L, Gain A.K, **Zhang M**, Huang X. Nano-Plasticity of Fused Silica under Coupled Thermo-Mechanical Loading. Materials Science & Engineering A. (在审，IF=7.0，对应学术论文第5章)

[3]**Zhang M,** Li Z, Zhang L. Subsurface damage evolution and material removal mechanism of monocrystalline silicon in ultrasonic vibration-assisted scratching: An experimental study. (待投稿，对应学位论文第4章)

申请及已获得的专利

[1]章亮炽, Asit Kumar Gain，张慕禹. 一种机械变形下表征非晶材料致密化的方法及系统（发明专利，2025，在审）



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